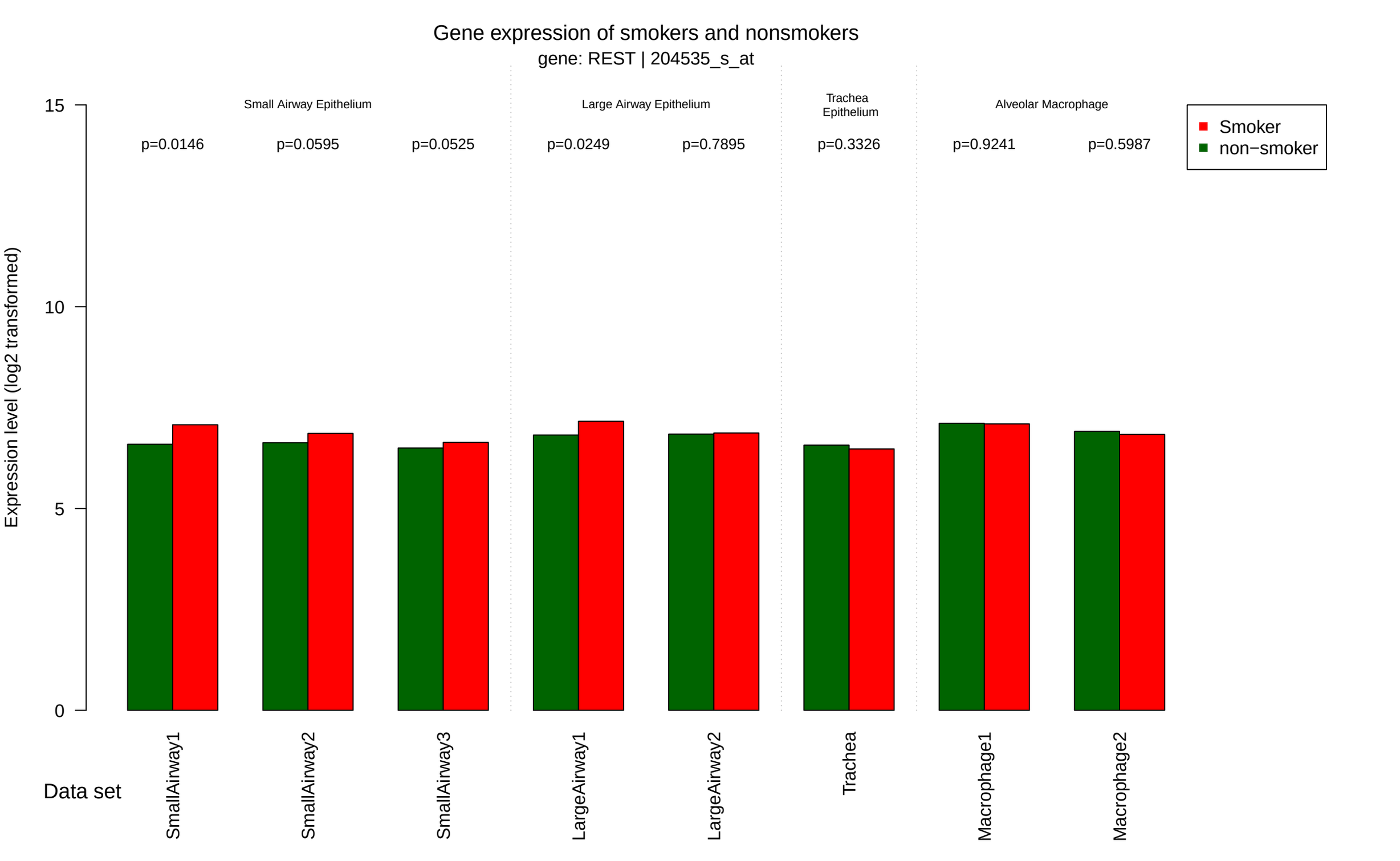
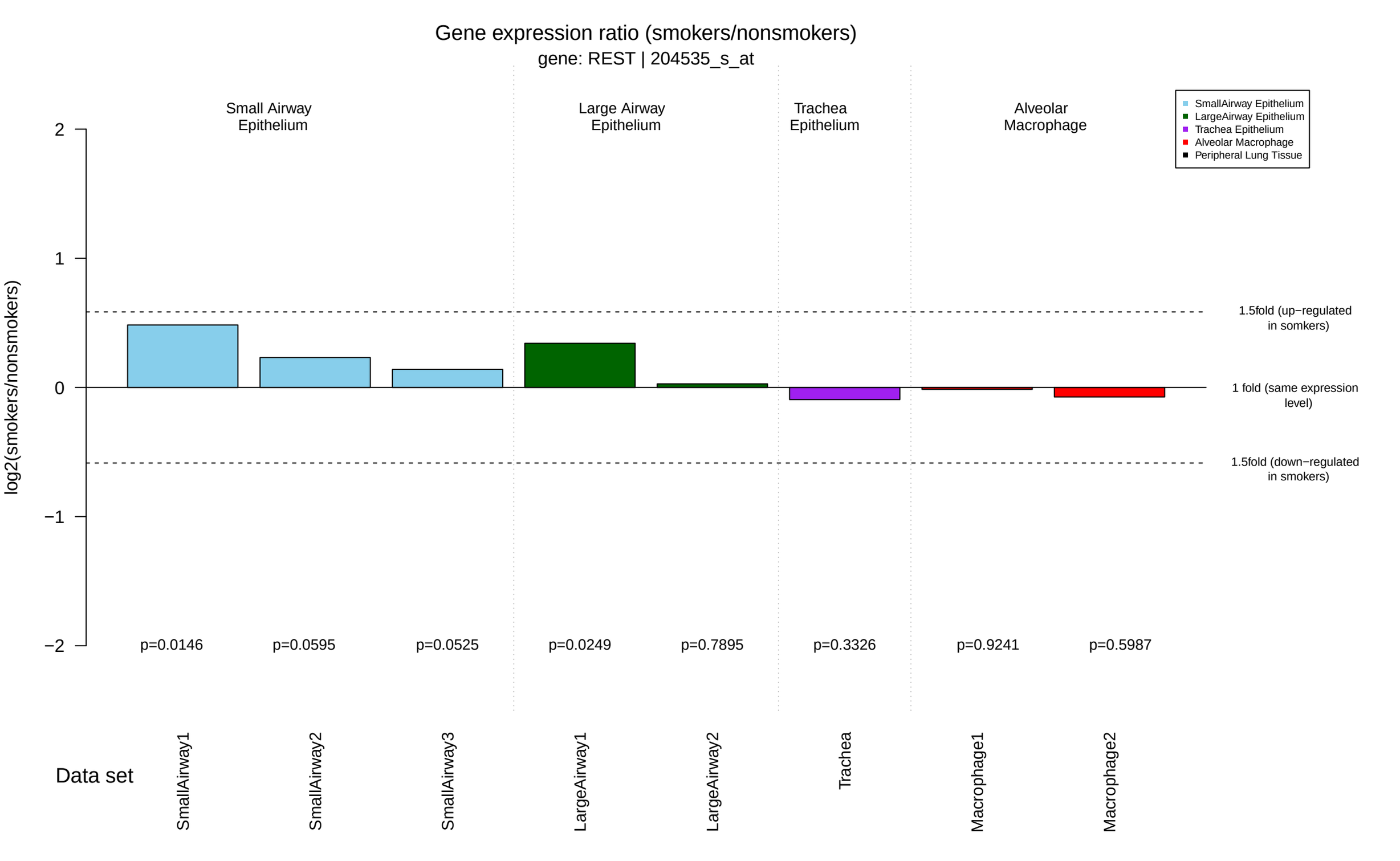
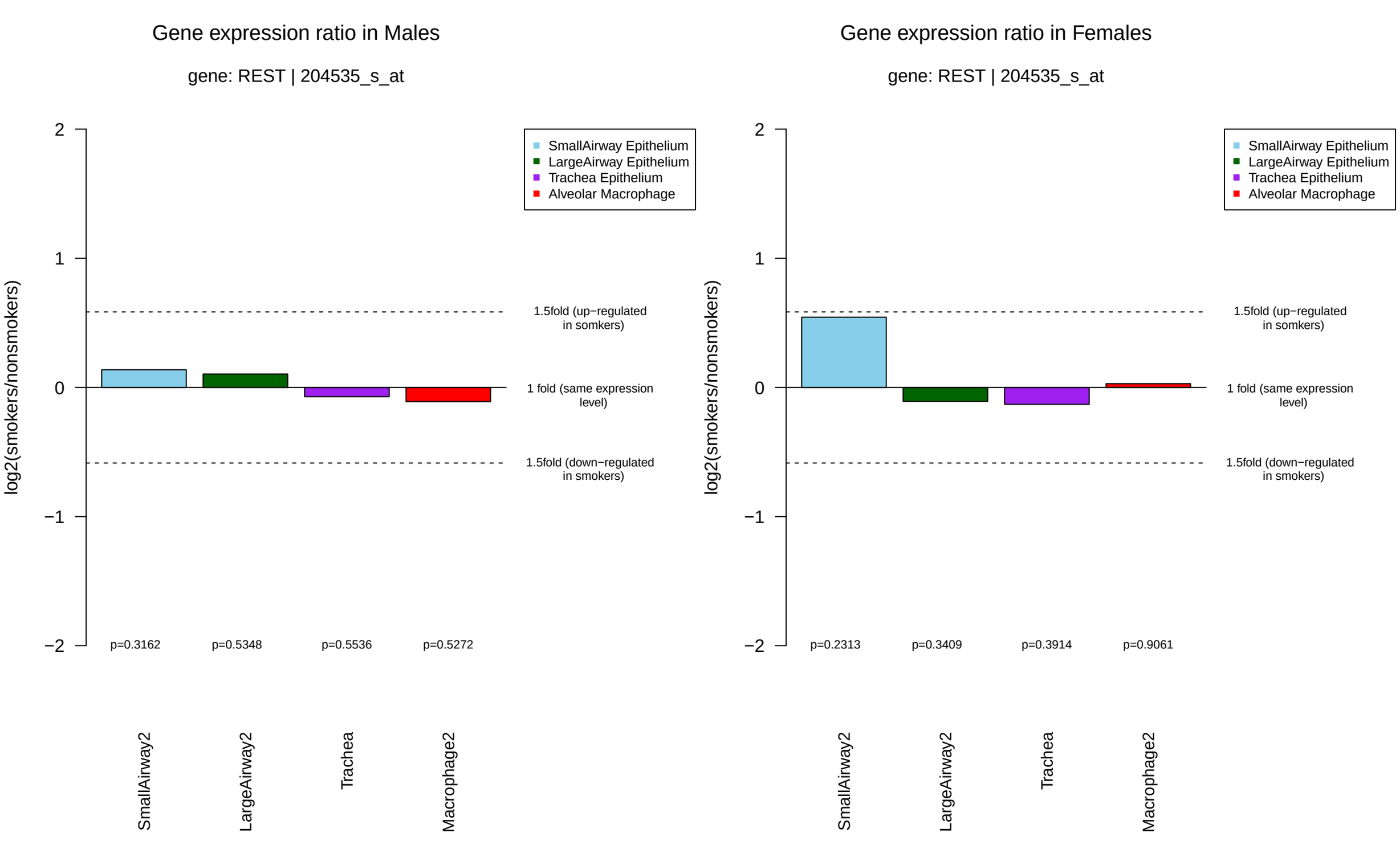
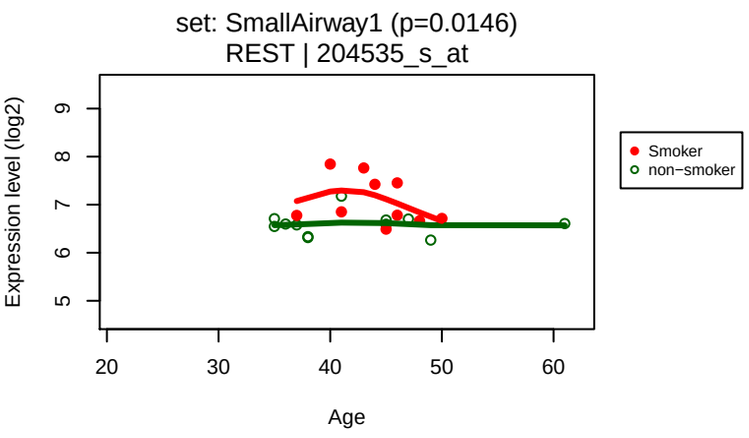
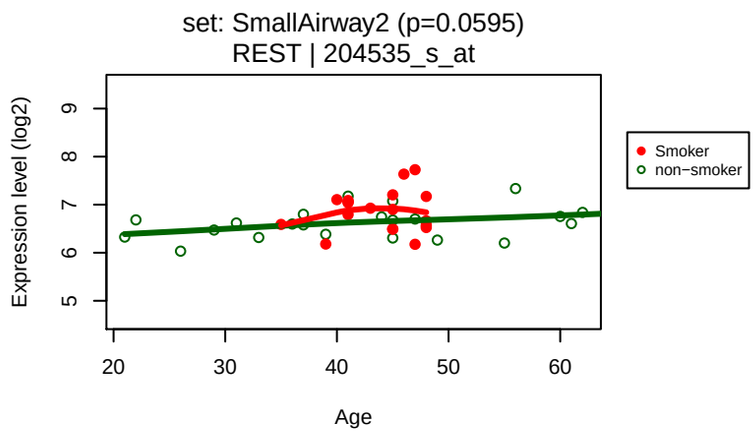
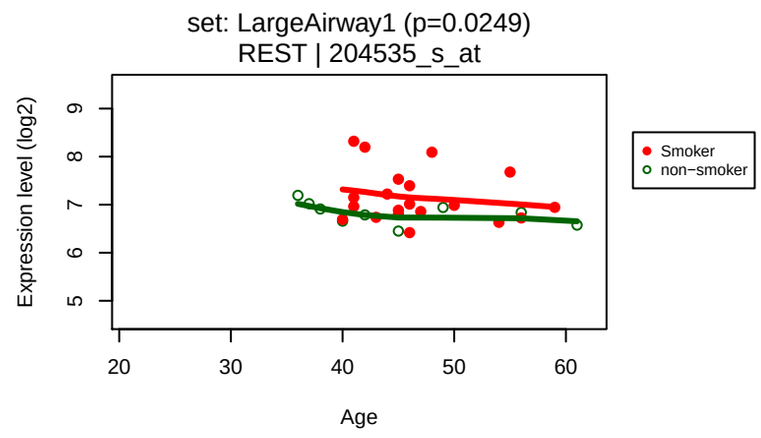

Gene expression of smokers and nonsmokers
gene: REST | 204535_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0146
p=0.0595
p=0.0525
p=0.0249
p=0.7895
p=0.3326
p=0.9241
p=0.5987
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

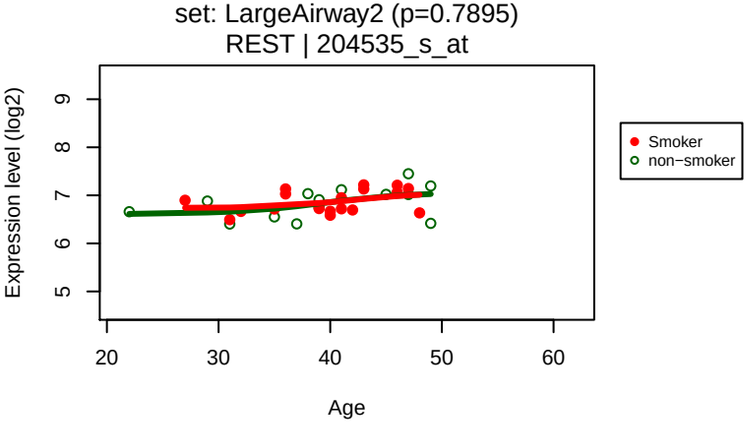

Gene expression ratio (smokers/nonsmokers)
gene: REST | 204535_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0146
p=0.0595
p=0.0525
p=0.0249
p=0.7895
p=0.3326
p=0.9241
p=0.5987
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

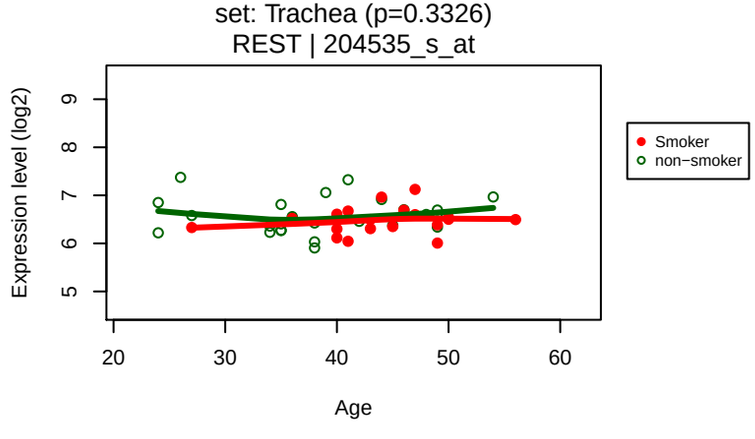

Gene expression ratio in Males
gene: REST | 204535_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3162
p=0.5348
p=0.5536
p=0.5272
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: REST | 204535_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2313
p=0.3409
p=0.3914
p=0.9061
SmallAirway2
LargeAirway2
Trachea
Macrophage2

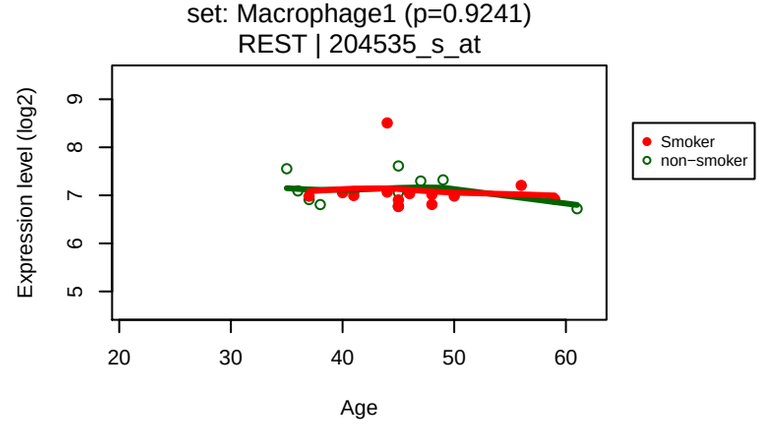

set: SmallAirway1 (p=0.0146)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker

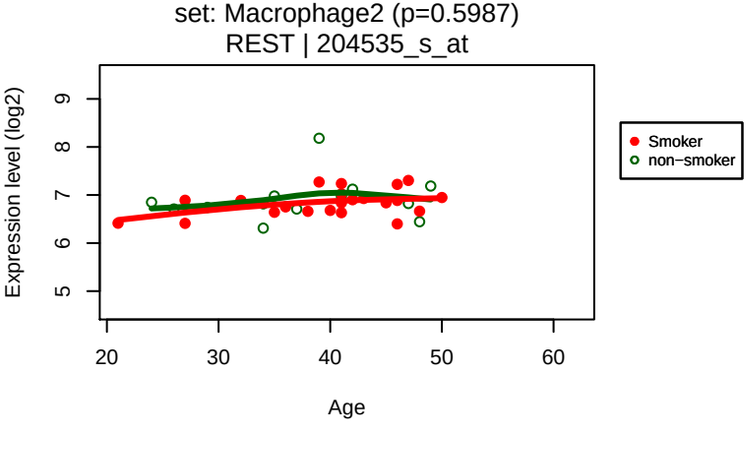

set: SmallAirway2 (p=0.0595)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker

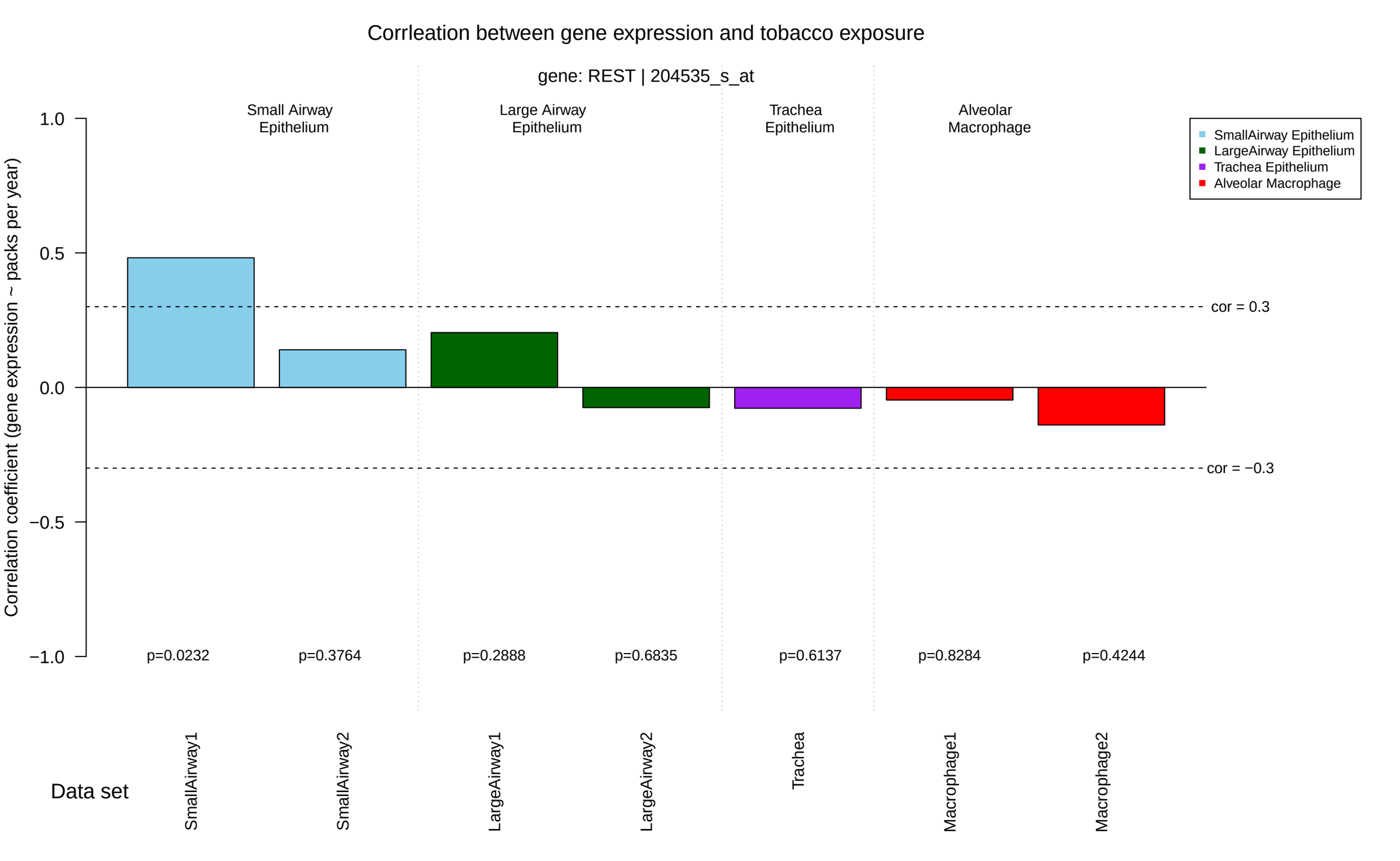

set: LargeAirway1 (p=0.0249)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7895)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3326)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9241)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5987)
REST | 204535_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: REST | 204535_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0232
p=0.3764
p=0.2888
p=0.6835
p=0.6137
p=0.8284
p=0.4244
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2